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P(TE) Analysis

As a reminder, Wendy's central question has been, "**which infected humans transmit to mosquitoes?**" P(TE) estimates, "how likely is this infected human the source of an infected mosquito?" The data contain 3,970 possible human-to-mosquito links (edges) from 576 human infections and 189 infected mosquitoes.

Given this, my initial OLS chart tried to identify unexpectedly high (inferred) transmitters, visualized as the highest residuals or difference between each human infection's summed P(TE) and predicted sum based on:

- parasite density (amount in human blood sample)
- symptoms (vs. asymptomatic)
- MOI (genetic complexity based on distinct parasite strains in human blood sample)
- human age
- household mosquito abundance

While I chose these predictors to control for biology and opportunity, JP helped me realize the summed P(TE) outcome was the issue, strongly influenced by how many opportunities each infection had to accumulate P(TE). Specifically, the 58 "surprising" infections (defined as top 10% of the total 576) had a median of 24 edges vs. 2 overall, and the number of links was strongly correlated with summed P(TE): r=0.75.

As a first fix, I replaced total P(TE) with average P(TE) per possible link as the opportunity-adjusted outcome, which nearly eliminated the relationship with edge count (r=0.04) and notably changed the focal set (only 8 of the original 58 infections remained surprising). Also, the new model had R^2=0.03, although this is not directly comparable with the original R^2=0.45 because the original model predicted log(1+summed P(TE)) while the new model predicted average P(TE)). Interestingly, MOI remained positively associated with average P(TE), while household mosquito abundance no longer showed a consistent relationship.

However, I noticed that many infections only had one possible link, so there might be substantial differences in precision. Upon exploration, I found that more than half only have 1-2 possible links. Across 576 human infections:

- 220 have exactly 1 link (38%)
- 88 have exactly 2 links (15%)
- **138 have at least 10 links** (24%)
- **80 have at least 20 links**
- Mean: 6.9 links
- Median: 2 links
- Maximum: 36 links

Instead of restricting the primary analysis to infections with stable averages, which could introduce selection bias by disproportionately analyzing infections from high-mosquito, unusually well-sampled environments, I retained all 576 infections and used those with >=5 and >=10 links as sensitivity checks to see whether sparse infections were driving the results:

Test	Infections	Surprising	R^2	Overlap with original 58
All links	576	58	.033	8
>= 5 links	200	20	.216	17
>= 10 links	138	14	.286	13

The better-supported sets were considerably more consistent with each other: 13 of the 14 surprising infections at $\geq$10 links were also surprising at $\geq$5, and 7 infections were surprising across all three specifications.

Conclusion: At this point, while the data support broader population-level transmission analyses, I thought it may not support reliable individual-level "surprising transmitter" ranking due to the sparse/uneven observation. Wendy voiced these concerns early on:

- Source ambiguity: each infected mosquito has a median of roughly 13 nonzero-P(TE) plausible human sources
- Incomplete observation: a person with low or zero inferred transmission may simply have had their transmitting mosquitoes not collected
- Genotype ambiguity: one person can carry multiple parasite strains and one mosquito can contain multiple strains
- Timing: persistent infections make temporal source attribution less certain

- Dependence: the 3,970 possible links arise from 576 human infections (199 people across 37 households) and 189 infected mosquito abdomens

However, as a final check, I split the analysis by CSP vs. AMA, the two parasite genetic markers the original authors used to independently calculate two versions of P(TE). For context, Sumner et al.'s main analysis emphasized CSP, which my original analysis also used for that reason, with AMA as a parallel analysis. If the same high-surprise infections emerge under both markets, that would strengthen the individual-level signal, and if they differ substantially, it would quantify how sensitive the surprise ranking is to genetic source attribution. Here are the results:

- All 576 infections: CSP vs AMA overlap = **10** of 58
- $\geq$5 links: overlap = **1** of 20
- $\geq$10 links: overlap = **2** of 14
- Only **1** infection is surprising under both CSP and AMA at both $\geq$5 and $\geq$10 links
- The residual correlations are essentially zero: 0.068, -0.023, and ~0.000

In other words, the same infections generally do not look unusually high-transmitting when switching from CSP-based to AMA-based P(TE). The individual-level surprise ranking therefore appears highly sensitive to which genetic marker is used to infer the human-to-mosquito transmission links. Even though individual surprising infections were highly unstable, MOI remained positively associated with average P(TE) across both markers and all three specifications, indicating that **infections with higher MOI tend to have higher average P(TE)**, after adjusting for the other covariates.

Review: Sumner et al. (2021) came closest to directly addressing human-to-mosquito transmission by constructing P(TE). Their main population-level conclusion was that 94.6% of mosquito infections were estimated to originate from asymptomatic human infections, although asymptomatic infections were extremely common and represented roughly 73-97% of human infections depending on the month. Markwalter et al. (2024) also attempted to find an individual-level conclusion, but had too few successfully infected reared mosquitoes to investigate reliably. The paper therefore pivoted to investigating biting opportunities by asking, "Who did mosquitoes actually bite?" based on blood meal matching, much closer to an observed event with ultimately 720 mosquito-human matches. They found biting was extremely concentrated with 20% of people receiving 86% of observed bites, with higher biting among males, ages 5-15, already-infected people, and non-net-users. Restricting to infected humans, they asked, "Among people capable of transmitting malaria, who receives the bites that could potentially infect mosquitoes?” They estimated that infected boys ages 5–15 received about 50% of potentially infectious bites. So, if infected Person A gets bitten 20 times, and infected Person B gets bitten 5 times, they found that Person A had more *opportunities* to infect mosquitoes (described as potential contribution to onward transmission, not confirmed human-to-mosquito transmission). **Unanswered:** However, it does not answer whether Person A *actually infected* 10 mosquitoes while Person B infected 0, or vice versa.